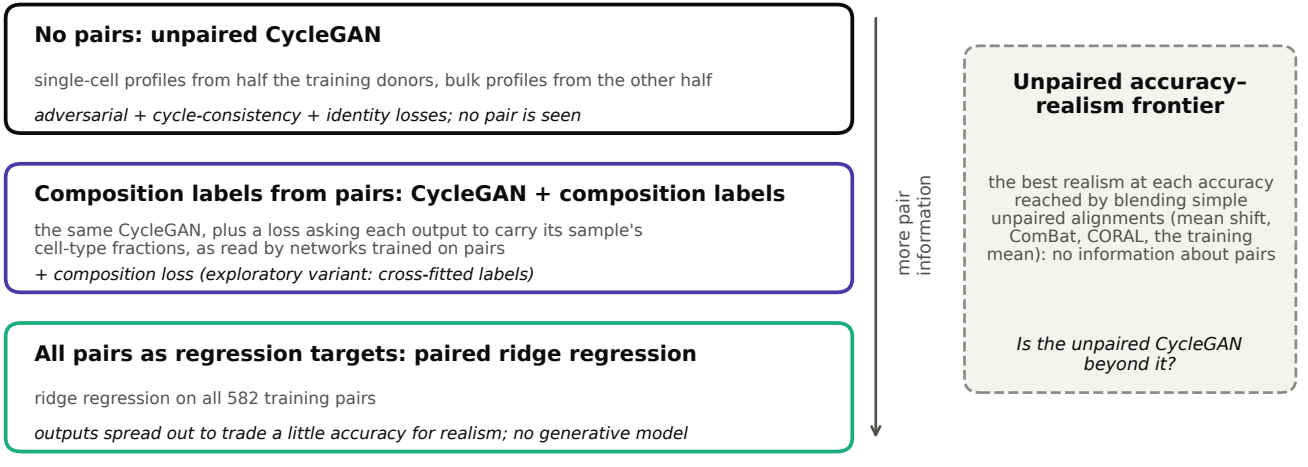
No pairs: unpaired CycleGAN
single-cell profiles from half the training donors, bulk profiles from the other half
adversarial + cycle-consistency + identity losses; no pair is seen
Composition labels from pairs: CycleGAN + composition labels
the same CycleGAN, plus a loss asking each output to carry its sample's cell-type fractions, as read by networks trained on pairs
+ composition loss (exploratory variant: cross-fitted labels)
All pairs as regression targets: paired ridge regression
ridge regression on all 582 training pairs
outputs spread out to trade a little accuracy for realism; no generative model
more pair information
Unpaired accuracy–realism frontier
the best realism at each accuracy reached by blending simple unpaired alignments (mean shift, ComBat, CORAL, the training mean): no information about pairs
Is the unpaired CycleGAN beyond it?